# SILDENAFIL

Never with nitrates

Take 1 h before sex

**CHEMICAL NAME** 5-[2-ethoxy-5-(4-methylpiperazin-1-ylsulfonyl)phenyl]-1-methyl-3-propyl-1,6-dihydro-7H-pyrazolo[4,3-d]pyrimidin-7-one (citrate)

**CLASS** Phosphodiesterase type 5 (PDE-5) inhibitor

**TRADE NAMES** Viagra · Revatio (pulmonary hypertension) · Vigrande

## THERAPEUTIC USES / INDICATIONS

- Erectile dysfunction.
- Pulmonary arterial hypertension (adults and children).
- Raynaud's phenomenon (off-label); high-altitude pulmonary oedema (off-label).

## ADMINISTRATION

- **PO** ED: about 1 h before sexual activity (delayed by fatty meals).
- **PAH:** regular dosing three times daily; IV form available.
- Sexual stimulation is required for effect.

## DOSAGES

- **ED:** 50 mg as needed (range 25–100 mg); max once daily.
- **Elderly/hepatic/renal impairment, CYP3A4 inhibitors:** start 25 mg.
- **PAH:** 20 mg three times daily (child weight-based).
- **IV (PAH):** 10 mg three times daily.

## PHARMACOKINETICS

- **Onset:** 30–60 min. **Duration:** ~4 h.
- Metabolised in liver (CYP3A4 major) to active N-desmethyl metabolite.
- Excreted in faeces. **t½:** 3–5 h.

## NURSING MANAGEMENT

- Check nitrate use; document clearly (inform emergency staff).
- PAH: monitor BP, $SpO_2$, exercise tolerance.
- Cardiac risk assessment before ED use.

## PHARMACODYNAMICS

- Inhibits **PDE-5** → ↑ cGMP in corpus cavernosum and pulmonary vasculature → smooth-muscle relaxation, vasodilation.
- Also weakly inhibits PDE-6 in retina → blue vision.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Headache, flushing.
- Dyspepsia, nasal congestion.
- Visual changes (blue tinge), dizziness.

### ADVERSE EFFECTS (SERIOUS)

- Severe hypotension with nitrates; potentially fatal.
- Priapism (>4 h) — emergency.
- NAION (sudden vision loss); sudden hearing loss.
- MI, arrhythmias in patients with cardiac disease.

## CONTRAINDICATIONS & CAUTIONS

- Nitrates (GTN, isosorbide) or nitric oxide donors; riociguat.
- Severe cardiovascular disease where sexual activity inadvisable; recent stroke/MI; hypotension <90/50.
- Hereditary retinal disorders (retinitis pigmentosa); previous NAION.
- Severe hepatic impairment.

### KEY INTERACTIONS

- **Nitrates:** contraindicated (wait 24 h after sildenafil).
- **α-blockers:** postural hypotension (separate dosing).
- **CYP3A4 inhibitors (ritonavir, clarithromycin):** ↑ levels.
- **Grapefruit:** ↑ levels.

## PATIENT EDUCATION

- Never use with GTN or other nitrates; tell emergency staff you have taken it.
- Seek urgent help if erection lasts >4 hours or sudden vision/hearing loss.
- Avoid heavy meals and alcohol before use.

**KEY POINTS**

- PDE-5 inhibitor — ED and PAH.
- Never with nitrates.
- ED 50 mg 1 h before.
- Priapism >4 h = emergency.